(Table 27, contd)

| Wheel Loaders, Integrated Toolcarriers, Wheeled Dozers, and Compactors Lubricant Viscosities for Ambient Temperatures | | | | | | |
|---|---|---|---|---|---|---|
| **Compartment or System** | **Oil Type and Performance Requirements** | **Oil Viscosities** | **°C** | | **°F** | |
| | | | **Min** | **Max** | **Min** | **Max** |
| **Drive Axles for** 994 Wheel Loader | **Cat TDTO** | SAE 10W | −25 | 0 | −13 | 32 |
| | | SAE 30 | −20 | 20 | −4 | 68 |
| | | SAE 50[5] | −10 | 43 | 14 | 110 |
| | **Cat TDTO-TMS** | Multi-Grade | −25 | 22 | −13 | 72 |

(1) Factory fill oil for standard configuration machines.
(2) Cat SATO is factory fill for APT and for Power Shift Transmission of M-Series 950-982 medium wheel loaders.
(3) Cat TDTO SAE 30 is factory fill for standard configuration Medium Wheel Loaders (950-980) and Medium Wheel Dozer (824).
(4) Cat TDTO SAE 50 is factory fill for standard configuration Small Wheel Dozer (814) and Medium Compactors (815-826).
(5) Dealer service fill for standard configuration machines.

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## Special Applications for 906, 907, 908

Table 28

| Compact Wheel Loaders Lubricant Viscosities for Ambient Temperatures | | | | | | |
|---|---|---|---|---|---|---|
| Compartment or System | Oil Type and Performance Requirements | Oil Viscosities | °C | | °F | |
| | | | Min | Max | Min | Max |
| **Brakes on Standard Drive** | Cat HYDO Advanced 10 | SAE 10W | −20 | 40 | −4 | 104 |

Table 29

| Special Additive Requirement for Axles on Wheel Loaders, IT machines, Wheel Dozers, and Compactors | |
|---|---|
| Required Volume of 1U-9891 Oil Additive or 185-4771 Oil Additive for axles that have Wet Disc, Axle Shaft Speed Brakes (1) | |
| **Machine Model** | **Number of bottles per axle** |
| All Compact and Small (910-930) Wheel Loaders<br>IT28-IT62 Integrated Tool Carriers<br>938-962 Wheel Loaders, except 938K and 938M(2) | 0.5 |
| 966F, 970F, 966G, and 972G Wheel Loaders<br>815 and 816 Compactors<br>814 Wheel Dozers | 1.0 |
| 980F, 980G, and 980H Wheel Loaders<br>824G and 824H Wheel Dozers<br>825G, 825H, 826G, and 826H Compactors | 1.5 |
| 988FII, 988G, and 988H Wheel Loaders<br>834B, 834G, and 834H Wheel Dozers<br>836, 836G, and 836H Compactors | 3 |
| 990 Series II Wheel Loaders<br>844 Wheel Dozers | 4 |
| 992G Wheel Loaders<br>854G Wheel Dozers<br>854K Wheel Dozers | 5 |
| 993K Wheel Loaders | 7 |

(1) Do not use 1U-9891 Oil Additive in brake compartments when the service brake is also used as the parking brake unless specifically recommended by Caterpillar.
(2) Do not use this additive with the 938K and 938M wheel loaders.

# Wheel Tractor-Scrapers

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models:

- 611 through 657
- TS180 through TS225

For hydraulic systems that are not given in Table 30 , refer to the hydraulic system recommendations given in Table 8 .

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Table 30

| Wheel Tractor-Scrapers Lubricant Viscosities for Ambient Temperatures | | | | | | |
|---|---|---|---|---|---|---|
| **Compartment or System** | **Oil Type and Performance Requirements** | **Oil Viscosities** | **°C** | | **°F** | |
| | | | **Min** | **Max** | **Min** | **Max** |
| **Transmission** | **Cat TDTO Cold Weather** | SAE 0W-20 | −40 | 0 | −40 | 32 |
| | **Cat TDTO** | SAE 10W | −20 | 10 | −4 | 50 |
| | | SAE 30(1) | 0 | 50 | 32 | 122 |
| | **Cat TDTO-TMS** | Multi-Grade | −15 | 50 | 5 | 122 |
| **Non-driven Scraper Wheels except for 613G(2)** | **Cat FDAO** | SAE 60(1) | −7 | 50 | 19 | 122 |
| | Cat TDTO | SAE 30 | −25 | 15 | −13 | 59 |
| | **Cat FDAO SYN** | Multi-Grade | −15 | 50 | 5 | 122 |
| | **Cat TDTO-TMS** | Multi-Grade | −35 | 15 | −31 | 122 |
| **Nondriven Scraper Wheels for 613G** | **Cat Utility Grease** | NLGI 2 | −30 | 40 | −22 | 104 |
| **Nondriven Scraper Wheels (Towed Scrapers)** | Cat Synthetic GO | SAE 75W-140 | −30 | 45 | −22 | 113 |
| | Cat GO (Gear Oil) | SAE 80W-90(1) | −20 | 40 | −4 | 104 |
| | | SAE 85W-140 | −10 | 50 | 14 | 122 |
| | API GL-5 gear oil | SAE 75W-90 | −30 | 40 | −22 | 104 |
| | | SAE 90 | 0 | 40 | 32 | 104 |
| **Differentials and Final Drives, Auger Lower Bearings, Auger, and Elevator Speed Reducers** | **Cat Synthetic GO** | SAE 75W-140 | −30 | 45 | −22 | 113 |
| | Cat GO (Gear Oil) | SAE 80W-90(1) | −20 | 40 | −4 | 104 |
| | | SAE 85W-140 | −10 | 50 | 14 | 122 |
| | API GL-5 gear oil | SAE 75W-90 | −30 | 40 | −22 | 104 |
| | | SAE 90 | 0 | 40 | 32 | 104 |
| **Brake Circulation for 620H Series** | **Cat TDTO Cold Weather** | SAE 0W-20 | -40 | 40 | -40 | 104 |
| | **Cat TDTO** | SAE 10W(1) | -20 | 50 | -4 | 122 |
| **Elevator Lower Rollers** | Cat DEO Cold Weather | SAE 0W-40 | −40 | 40 | −40 | 104 |
| | Cat DEO-ULS<br>Cat DEO | SAE 15W-40 | −10 | 50 | 14 | 122 |

(1) Factory fill oil for standard configuration machines.
(2) Non-driven scraper wheels (except for 613G) can also use the gear oils listed under Differentials and Final Drives in this table.

## Petroleum Transmissions

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models:

- TH31-E61 through TH55-E90
- CX31-P600
- CX35-P800

For hydraulic systems that are not given in Table 31 , refer to the hydraulic system recommendations given in Table 8 .

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Table 31

| Petroleum Transmissions Lubricant Viscosities for Ambient Temperatures | | | | | | |
|---|---|---|---|---|---|---|
| Compartment or System | Oil Type and Performance Requirements | Oil Viscosities | °C | | °F | |
| | | | Min | Max | Min | Max |
| Petroleum Transmissions TH48-E70, TH48-E80, TH55-E70, and TH55-E90 | Cat TDTO Cold Weather | SAE 0W-20 | −40 | 10 | −40 | 50 |
| | Cat TDTO | SAE 10W | −20 | 10 | −4 | 50 |
| | | SAE 30(1) | 0 | 50 | 32 | 122 |
| | Cat TDTO-TMS | Multi-Grade | −10 | 50 | 14 | 122 |
| Petroleum Transmissions TH31-E61and TH35-E81 | Cat TDTO Cold Weather | SAE 0W-20 | −40 | 22 | −40 | 72 |
| | Cat TDTO | SAE 10W | −20 | 22 | −4 | 72 |
| | | SAE 30(1) | 10 | 50 | 50 | 122 |
| | Cat TDTO-TMS | Multi-Grade | −15 | 50 | 5 | 122 |

(1) Factory fill oil for standard configuration machines.

## On-Highway Transmissions

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes the following models:

- CX28, CX31, and CX35

Cat CX Series transmissions are used in both on and off-highway.

Typical on-highway applications include: line haul, pick up and delivery, beverage delivery, public service dump, emergency vehicles, and recreational vehicles with living accommodations.

Typical off-highway applications include: dump trucks, transit mixers, refuse trucks, all wheel drive public utility trucks, yard spotters, concrete pumpers, heavy equipment transport, and specialty PTO applications.

Fluid change intervals for Standard Cat CX Series transmissions and Standard and Retarder Cat CX Series transmissions are different. Refer to the Operation and Maintenance Manual for your transmission or consult your Cat dealer for details.

Fluid change intervals for severe applications are shorter than fluid change intervals for general applications. Refer to the Operation and Maintenance Manual for your transmission or consult your Cat dealer for details.

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Table 32

| On-Highway Transmissions Lubricant Viscosities for Ambient Temperatures | | | | | | |
|---|---|---|---|---|---|---|
| **Compartment or System** | **Oil Type and Performance Requirements** | **Oil Viscosities** | **°C** | | **°F** | |
| | | | **Min** | **Max** | **Min** | **Max** |
| **On-Highway Transmissions CX28, CX31, and CX35** | **Cat ATF-HD2[1][2]** | Multi-Grade Synthetic | −23 | 45 | −10 | 113 |

(1) For temperatures lower than −23° C (−10° F), warm up the engine for 20 minutes or use an appropriate transmission oil heater.
(2) Cat AT-1, Dexron III, Dexron IV, and Dexron VI specifications define the minimum requirements for alternative commercial oils for use in Cat CX series transmissions.

## Special Applications

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the special applications that apply to Cat equipment.

Table 33

| Special Applications Lubricant Viscosities for Ambient Temperatures | | | | | | |
|---|---|---|---|---|---|---|
| **Compartment or System** | **Oil Type and Performance Requirements** | **Oil Viscosities** | **°C** | | **°F** | |
| | | | **Min** | **Max** | **Min** | **Max** |
| **Starting Engine** | **SH SJ SL** | SAE 0W-20 | −40 | 40 | −40 | 104 |
| | | SAE 0W-30 | −40 | 40 | −40 | 104 |
| | | SAE 5W-20 | −30 | 10 | −22 | 50 |
| | | SAE10W | −20 | 50 | −4 | 122 |
| **Starting Engine Transmission** | **Cat TDTO Cold Weather** | SAE 0W-20 | −40 | 10 | −40 | 50 |
| | **Cat DEO Cold Weather** | SAE 0W-40 | −40 | 40 | −40 | 104 |
| | **Cat DEO-ULS SYN** | SAE 5W-40 | −40 | 50 | −40 | 122 |
| | **Cat TO-4** | SAE 5W-20 | −30 | 10 | −22 | 50 |
| | **Cat TDTO** | SAE 10W | −30 | 20 | −22 | 68 |
| | | SAE 30 | −10 | 25 | 14 | 77 |
| **Variable Pitch Fan** | Cat DEO Cold Weather | SAE 0W-40 | −40 | 40 | −40 | 104 |
| | **Cat DEO-ULS SYN** | SAE 5W-40 | −40 | 50 | −40 | 122 |
| **Hydraulic Hammer** | Cat HYDO Advanced 10 | SAE 10W | −20 | 50 | −4 | 122 |
| | Cat HYDO Advanced 20 | SAE 20 | -5 | 45 | 23 | 113 |
| | Cat HYDO Advanced 30 | SAE 30 | 10 | 50 | 50 | 122 |
| | Cat BIO HYDO Advanced | “ISO 46” Multi-Grade | −40 | 40 | −4 | 104 |
| | Cat ECF-1-a, Cat ECF-2, Cat ECF-3, Cat TO-4 | SAE 0W-30 | −40 | 10 | −40 | 50 |
| | Cat DEO-ULS Cat DEO | SAE 15W-40 | −15 | 50 | 5 | 122 |
| | Cat MTO | SAE10W-30 | −20 | 40 | −4 | 104 |
| | Cat TDTO-TMS | Multi-grade | −15 | 50 | 5 | 122 |
| **Bar and Chain Oil** | Cat DEO Cold Weather | SAE 0W-40 | −40 | 40 | −40 | 104 |
| | Cat DEO-ULS **Cat DEO** | SAE 10W-30 | −18 | 40 | 0 | 104 |
| | | SAE 15W-40 | −10 | 50 | 14 | 122 |

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# Expanded Mining Fluids

**SMCS Code:** 1000; 7000; 7581

## Expanded Mining Fluids Recommendations

Refer to this Special Publication, “General Information for Lubricants” for important lubricant information.

For the engine oil recommendations of all Expanded Mining Products, refer to the “Cat Diesel Engines Lubricant Viscosities for Ambient Temperatures” table in this Special Publication, “Lubricant Viscosities”.

The fluids recommended for Expanded Mining Products follow the same guidelines, recommendations, and information given for Cat machines in this Special Publication.

Refer to the machine Owning and Operating Manual, and to your Cat dealer for fluids information and availability.

When the oil type (or brand) is changed in a machine component or an engine, ensure that the system is flushed properly before changing the oil. This process reduces risk of mixing oil types that can be incompatible and that can reduce the effectiveness or cause other negative impacts. Caterpillar offers multiple publications to guide system flushing. Refer to Cat Media Information Center, your machine Operation and Maintenance Manual, or your dealer for information.

## Surface Mining Products

### Draglines

Refer to this Special Publication, “General Information for Lubricants” for important lubricant information.

This section includes, but is not limited to the following models:

- 8200 and 8750

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Table 34

| Dragline Fluid Recommendations for Ambient Temperatures | | | | | | |
|---|---|---|---|---|---|---|
| **Compartment or System** | **Oil Type and Performance Requirements** | **Oil Viscosities** | **°C** | | **°F** | |
| | | | **Min** | **Max** | **Min** | **Max** |
| **Open Gear Lube Reservoir** | **Cat Open Gear Lubricant** | **OGL Cold Weather** | **-40** | **10** | **-40** | **50** |
| | | **OGL Warm Weather** | **-7** | **49** | **20** | **120** |
| **Multi-Purpose Grease Reservoir (Rolling Element Bearings)** | **Cat Extreme Application Grease** | **NLGI 0** | **No Min** | **-12** | **No Min** | **10** |
| | | **NLGI 1** | **-18** | **16** | **0** | **61** |
| | | **NLGI 2** | **0** | **16** | **32** | **61** |
| | **Cat Extreme Application Grease-Desert** | **NLGI 2** | **0** | **40** | **32** | **104** |
| **Parallel Shaft Swing Gearcases, Hoist/Drag Gearcase and Propel Gearcases** | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| **Planetary Hoist Helper Drive Gearcase, Swing Planetary Gearcase** | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| **Walking Cam Mechanism for** models 7850, 8200and 8750 | Cat Open Gear Lubricant | OGL Cold Weather | -40 | 10 | -40 | 50 |
| | | OGL Warm Weather | -7 | 49 | 20 | 120 |
| **Monigham Style Walking Mechanism for** Bucyrus models 1370W and 1570W | Refer to OEM Recommendations | | | | | |
| **Cam & Slide Walking Mechanism for** Bucyrus model 2570W | Refer to OEM Recommendations | | | | | |
| **Couplings** | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| **Air Compressors - All Brands** | Refer to OEM Recommendations | | | | | |

## Rotary and Track Drills

Refer to this Special Publication, “General Information for Lubricants” for important lubricant information.

This section includes, but is not limited to the following models:

Rotary Drills

- Diesel MD6240 through MD6750
- Electric MD6640 through MD6750

Track Drills

- MD5050 through MD5150

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Table 35

| Diesel Rotary Drills Fluid Recommendations for Ambient Temperatures [1] | | | | | | |
|---|---|---|---|---|---|---|
| **Compartment or System** | **Oil Type and Performance Requirements** | **Oil Viscosities** | **°C** | | **°F** | |
| | | | **Min** | **Max** | **Min** | **Max** |
| Hydraulic System (All drills except MD6200) Hydraulic System (Crane) | Cat HYDO Advanced | SAE 10W [2] | –20 | 40 | -4 | 113 |
| | | SAE 20 [3] | -5 | 45 | 23 | 113 |
| | | SAE 30 | 0 | 50 | 32 | 122 |
| | Cat TDTO Cold Weather | SAE 0W-20 | -40 | 40 | -40 | 104 |
| Hydraulic System (MD6200) | Cat HYDO Advanced 10 | SAE 10W | -20 | 35 | -4 | 95 |
| | Cat HYDO Advanced 20 | SAE 20W | -5 | 40 | 23 | 104 |
| | Cat HYDO Advanced 30 | SAE 30 | 0 | 50 | 32 | 122 |
| | Cat BIO HYDO Advanced | "ISO 46" Multi–Grade | -30 | 40 | -22 | 104 |
| | Cat MTO Advanced 10<br>Cat DEO-ULS<br>Cat DEO | SAE 10W-30 | -20 | 35 | -4 | 95 |
| | Cat DEO-ULS<br>Cat DEO | SAE 15W-40 | -10 | 53 | 5 | 127 |
| | Cat TDTO-TMS | Multi-Grade | -15 | 53 | 5 | 127 |
| | Cat DEO-ULS Syn | SAE 5W-40 | -20 | 53 | -22 | 127 |
| | Cat DEO Cold Weather | SAE 0W-40 | -40 | 35 | -40 | 95 |
| | Cat TDTO Cold Weather | SAE 0W-20 | -40 | 35 | -40 | 95 |
| Air Compressor | Cat ATF-HD2 | Multigrade Synthetic | -29 | 49 | -20 | 120 |
| | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| Pump Drive, Rotary Head, and Final Drive | Cat Synthetic GO | SAE 75W-140 | —30 | 45 | -22 | 113 |
| | Cat GO (Gear Oil) | SAE 80W-90 | -20 | 40 | -4 | 104 |
| | | SAE-85W-140 | -10 | 50 | 14 | 122 |
| Water Injection System Pump | Cat DEO-ULS (API CJ-4)<br>Cat DEO (API CI-4/CI-4 PLUS) | SAE 15W-40 | -10 | 40 | 14 | 122 |
| Washing System Pump | Refer to the machine Operation and Maintenance Manual and/or consult your Cat dealer | SAE 30 | –18 | 35 | 0 | 95 |
| Water Injection Pump | | ISO 100<br>SAE 30 | -18 | 35 | 0 | 95 |
| | | ISO 220<br>SAE 50 | 35 | 49 | 95 | 120 |
| | | ISO 68<br>SAE20 | -34 | 16 | -30 | 60 |
| | | ISO 150<br>SAE 40 | -18 | 35 | 0 | 95 |
| Mast Winch | | ISO 220 | -4 | 54 | 25 | 130 |
| | | SAE 50 | -12 | 38 | 10 | 100 |

(continued)

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(Table 35, contd)

| Diesel Rotary Drills Fluid Recommendations for Ambient Temperatures [1] | | | | | | |
|---|---|---|---|---|---|---|
| Compartment or System | Oil Type and Performance Requirements | Oil Viscosities | °C | | °F | |
| | | | Min | Max | Min | Max |
| Central Grease System, Sheaves, Bearings | Cat Extreme application Grease | Cold Weather, NLGI 0 | -40 | 35 | -40 | 95 |
| | | High Loads, NLGI 2 | —30 | 50 | -22 | 122 |
| | Cat Prime Application Grease | Medium Loads, NLGI 2 | -20 | 40 | -4 | 104 |
| | Cat Utility Grease | Low Loads, NLGI 2 | -30 | 25 | -4 | 77 |
| Undercarriage - Track Idlers and Track Rollers | Cat DEO (single grade) API CF | SAE 30 | -20 | 25 | -4 | 77 |
| | Cat DEO-ULS SYN | SAE 5W -40 | -35 | 40 | -31 | 104 |
| Cat Autolube System | Cat Extreme Application Grease | NLGI Grade 1 | -35 | 40 | -31 | 104 |
| | | NLGI Grade 2 | -30 | 50 | –22 | 122 |

[1] Additional fluids may be required. Refer to your machine OMM for details.
[2] Factory fill for hydraulic track.
[3] Factory fill for rotary drill and rotary blasthole.

Table 36

| Electric Rotary Drills Fluid Recommendations for Ambient Temperatures [1] | | | | | | |
|---|---|---|---|---|---|---|
| Compartment or System | Oil Type and Performance Requirements | Oil Viscosities | °C | | °F | |
| | | | Min | Max | Min | Max |
| Hydraulic System | Cat HYDO Advanced | SAE 10W [2] | –20 | 40 | -4 | 113 |
| | | SAE 20 [3] | -5 | 45 | 23 | 113 |
| | | SAE 30 | 0 | 50 | 32 | 122 |
| | Cat TDTO Cold Weather | SAE 0W-20 | -40 | 40 | -40 | 104 |
| Air Compressor | Cat ATF-HD2 | Multigrade Synthetic | -29 | 49 | -20 | 120 |
| | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| Pump Drive, Rotary Head, and Final Drive | Cat Synthetic GO | SAE 75W-140 | —30 | 45 | -22 | 113 |
| | Cat GO (Gear Oil) | SAE 80W-90 | -20 | 40 | -4 | 104 |
| | | SAE-85W-140 | -10 | 50 | 14 | 122 |
| Mast Winch | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| Mast Guide Rollers | Cat Utility Grease | Low Loads, NLGI 2 | -30 | 440 | -22 | 104 |
| Central Grease System, Sheaves, Bearings | Cat Extreme application Grease | Cold Weather, NLGI 0 | -40 | 35 | -40 | 95 |
| | | High Loads, NLGI 2 | —30 | 50 | -22 | 122 |
| | Cat Prime Application Grease | Medium Loads, NLGI 2 | -20 | 40 | -4 | 104 |

(continued)

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(Table 36, contd)

| Electric Rotary Drills Fluid Recommendations for Ambient Temperatures [1] | | | | | | |
|---|---|---|---|---|---|---|
| Compartment or System | Oil Type and Performance Requirements | Oil Viscosities | °C | | °F | |
| | | | Min | Max | Min | Max |
| | Cat Utility Grease | Low Loads, NLGI 2 | -30 | 25 | -4 | 77 |
| Undercarriage - Track Idlers and Track Rollers | Cat DEO (single grade) API CF | SAE 30 | -20 | 25 | -4 | 77 |
| | Cat DEO-ULS SYN | SAE 5W -40 | -35 | 40 | -31 | 104 |
| Cat Autolube System | Cat Extreme Application Grease | NLGI Grade 1 | -35 | 40 | -31 | 104 |
| | | NLGI Grade 2 | -30 | 50 | –22 | 122 |

(1) Additional fluids may be required. Refer to your machine OMM for details.
(2) Factory fill for hydraulic track.
(3) Factory fill for rotary drill and rotary blasthole.

Table 37

| Diesel and Electric Rotary Drills Grease Recommendations for Ambient Temperatures | | | | | | |
|---|---|---|---|---|---|---|
| Compartment or System | Oil Type and Performance Requirements | NLGI Grade | °C | | °F | |
| | | | Min | Max | Min | Max |
| External Lubrication Points | Cat Prime Application Grease | NLGI Grade 2 | –20 | 50 | -4 | 122 |
| | Cat Extreme Application Grease | NLGI Grade 1 | –20 | 50 | -4 | 122 |
| | | NLGI Grade 2 | -15 | 50 | 5 | 122 |
| | Cat Extreme Application Grease Grease-Arctic | NLGI Grade 0 | 0 | 50 | –58 | 122 |
| | Cat Extreme Application Grease Grease-Desert | NLGI Grade 2 | -40 | 40 | 14 | 104 |
| | Cat Utility Grease | NLGI Grade 2 | -29 | 40 | -20 | 104 |
| | Cat Ball Bearing Grease | NLGI Grade 2 | | 50 | | 122 |
| Drill Lubrication System | Refer to the machine Operation and Maintenance Manual and/or consult your Cat dealer | | | | | |

Table 38

| Track Drills Fluid Recommendations for Ambient Temperatures [1] | | | | | | |
|---|---|---|---|---|---|---|
| Compartment or System | Oil Type and Performance Requirements | Oil Viscosities | °C | | °F | |
| | | | Min | Max | Min | Max |
| **Hydraulic System (Percussion Drills)** | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| Final Drive, Drill Feed Drive | Cat Synthetic GO | SAE 75W-140 | -30 | 45 | -22 | 113 |
| | Cat GO (Gear Oil) | SAE 80W-90 [2] | -20 | 40 | -4 | 104 |
| | | SAE 85W-140 | -10 | 50 | 14 | 122 |
| **Central Grease System, Bearings** | Cat Extreme Application Grease | Cold Weather, NLGI 0 | -40 | 35 | -40 | 95 |
| | | High Loads, NLGI 2 | -30 | 50 | -22 | 122 |
| | Cat Prime Application Grease | Medium Loads, NLGI 2 | -20 | 40 | -4 | 104 |

(continued)

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(Table 38, contd)

| Track Drills Fluid Recommendations for Ambient Temperatures [1] | | | | | | |
|---|---|---|---|---|---|---|
| Compartment or System | Oil Type and Performance Requirements | Oil Viscosities | °C | | °F | |
| | | | Min | Max | Min | Max |
| | Cat Utility Grease | Low Loads, NLGI 2 | -30 | 40 | -22 | 104 |
| Undercarriage - Bearings, Idlers | Cat Prime Application Grease | Low Loads, NLGI 2 | -30 | 40 | -22 | 104 |
| | | Medium Loads, NLGI 2 | -20 | 40 | -4 | 104 |
| | Cat Extreme Application Grease | Cold Weather, NLGI 0 | -40 | 35 | -40 | 95 |
| | | High Loads, NLGI 2 | -30 | 50 | -22 | 122 |
| Percussion Drill Grease[3] | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| Thread Lubrication System | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| Air Compressors | Cat ATF-HD2 | Multigrade Synthetic | -29 | 49 | -20 | 120 |
| | Refer to OEM Recommendations or consult your Cat Dealer | | | | | |
| Cat Autolube System | Cat Extreme Application Grease | NLGI Grade 1 | -35 | 40 | -31 | 104 |
| | | NLGI Grade 2 | -30 | 50 | -22 | 122 |

(1) Additional fluids may be required. Refer to your machineOperation and Maintenance Manual for details.
(2) Typical factory fill oil for standard configuration machines.
(3) Factory fill for standard configuration machines is Special Rock Drill Grease YPG 55.

## Electric Rope Shovels

Refer to this Special Publication, “General Information for Lubricants” for important lubricant information.

This section includes, but is not limited to the following models:

- 7295, 7395, 7495, 7495HDand 7495HF

Table 39

| Electric Rope Shovels Fluid Recommendations for Ambient Temperatures | | | | | | |
|---|---|---|---|---|---|---|
| Compartment or System | Oil Type and Performance Requirements | Oil Viscosities | °C | | °F | |
| | | | Min | Max | Min | Max |
| Open Gear Lube Reservoir (Open Gears, Roller Paths, Bushings, Dipper Handle, Crowd Take-Up) A1, A2, B1, B2 Systems | Cat Open Gear Lubricant | OGL Cold Weather | -40 | 10 | -40 | 50 |
| | | OGL Warm Weather | -7 | 49 | 20 | 120 |
| Multi-Purpose Grease Reservoir (Rolling Element Bearings) C1, C2 Systems | Cat Extreme Application Grease | NLGI 0 | No Min | -12 | No Min | 10 |
| | | NLGI 1 | -18 | 16 | 0 | 61 |
| | | NLGI 2 | 0 | 16 | 32 | 61 |
| | Cat Extreme Application Grease-Desert | NLGI 2 | 0 | 40 | 32 | 104 |
| Electric Motor, Hoist Motor, Propel Motor, Crowd Motor and Swing Motor | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |

(continued)

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(Table 39, contd)

| Electric Rope Shovels Fluid Recommendations for Ambient Temperatures | | | | | | |
|---|---|---|---|---|---|---|
| **Compartment or System** | **Oil Type and Performance Requirements** | **Oil Viscosities** | **°C** | | **°F** | |
| | | | **Min** | **Max** | **Min** | **Max** |
| **Air Compressors** All Brands | Refer to OEM Recommendations | | | | | |
| **Couplings** | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| **HydraCrowd**(1) | HYDO Advanced 10 | 10W | 0 | 50 | 32 | 122 |
| | Cat TDTO Cold Weather | 0W-20 | -40 | 40 | -40 | 104 |
| **Crowd Take-up Power Unit and Boarding Stairs Hydraulic Units** | Cat TDTO Cold Weather | 0W-20 | -40 | 40 | -40 | 104 |
| **Hoist Gearcases, Swing Gearcases, Crowd Gearcases, Auxiliary Winch, and HydraCrowd Pump Drive Transmission Gearcase** except 7295 Shovels | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| **Hoist Gearcases, Swing Gearcases, Crowd Gearcases for** 7295 Shovels | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| **Propel Gearcases for** all models | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |

(1) For compatibility with contamination monitors, this oil must be clear to very light amber, with no dyes or dark color additives.

## Hydraulic Shovels

Refer to this Special Publication, “General Information for Lubricants” for important lubricant information.

This section includes, but is not limited to the following models:

- 6015 through 6090

Table 40

| Hydraulic Shovels Fluid Recommendations for Ambient Temperatures | | | | | | |
|---|---|---|---|---|---|---|
| **Compartment or System** | **Oil Type and Performance Requirements** | **Oil Viscosities** | **°C** | | **°F** | |
| | | | **Min** | **Max** | **Min** | **Max** |
| **Hydraulic System** | Cat HYDO Advanced 10 | SAE 10W | -15 | 45 | 5 | 113 |
| | Cat HYDO Advanced 20 | SAE 20W | -7 | 55 | 19 | 131 |
| | Cat HYDO Advanced 30 | SAE 30 | 5 | 55 | 41 | 131 |
| **Pump Distribution Gearbox, Travel Gearbox, and Swing Gearbox** | Cat Compactor Oil | Synthetic, ISO 220 | -35 | 50 | -31 | 122 |
| **Central Lubrication System (Bearings, Slew Rings, Track System) External Lubrication Points** | Cat Extreme Application Grease | NLGI 2 | 0 | 55 | 32 | 131 |
| | Cat Extreme Application Grease | NLGI 1 | -20 | 38 | -4 | 100 |
| | Cat Extreme Application Grease-Arctic | NLGI 0 | -45 | -12 | -49 | -10 |

(continued)

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(Table 40, contd)

| Hydraulic Shovels Fluid Recommendations for Ambient Temperatures | | | | | | |
|---|---|---|---|---|---|---|
| **Compartment or System** | **Oil Type and Performance Requirements** | **Oil Viscosities** | **°C** | | **°F** | |
| | | | **Min** | **Max** | **Min** | **Max** |
| **Electric Drive Motor Bearings** | Cat Utility Grease | NLGI 2 | -30 | 40 | -22 | 104 |
| **Idlers, Track rollers, and Support Rollers Except 6015B[1][2]** | Tribol 3020/1000–0<br>Cat FDAO | NLGI 0<br>SAE 60 | -40 | 120 | -40 | 248 |
| **Idlers, Track rollers, and Support Rollers for 6015B[1]** | Cat DEO | API CF<br>SAE 40 | -40 | 120 | -40 | 248 |

(1) Components are pre-lubricated at the factory with a lifetime lubricant. For inquiries on recommended fluid in the event of a component re-build, contact your Cat dealer.
(2) Oil type depends on the component type. Fixed axle rollers are Cat FDAO, all other components are Tibol.

## Unit Rig Mining Trucks

Refer to this Special Publication, “General Information for Lubricants” for important lubricant information.

This section includes, but is not limited to the following models:

- MT3300 through MT6300

Table 41

| Unit Rig Mining Trucks Fluid Recommendations for Ambient Temperatures[1] | | | | | | |
|---|---|---|---|---|---|---|
| **Compartment or System** | **Oil Type and Performance Requirements** | **Oil Viscosities** | **°C** | | **°F** | |
| | | | **Min** | **Max** | **Min** | **Max** |
| **Front Wheel Bearings (Grease Lubricated)** | Cat Extreme Application Grease | NLGI 2 | -20 | 40 | -4 | 104 |
| | Cat Prime Application Grease | NLGI 2 | -30 | 50 | -22 | 122 |
| **Front Wheel Bearings (Oil Lubricated)** | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |
| | Cat Compactor Oil | Synthetic, ISO 220 | -37 | 25 | -35 | 77 |
| **Hydraulic Systems** for MT4400D and MT5300D Models | Cat TDTO | SAE 10W | -20 | 50 | -4 | 122 |
| **Hydraulic Systems** for all models except D Series | Cat TDTO Cold Weather | SAE 0W-20 | -40 | 40 | -40 | 104 |
| | Cat HYDO Advanced | SAE 10W | -20 | 40 | -4 | 113 |
| | | SAE 20W | -5 | 45 | 23 | 113 |
| | | SAE 30 | 0 | 50 | 32 | 122 |
| **Final Drive / Wheel Motors** for MT4400D and MT5300D Models | Cat FDAO | SAE 60 | -10 | 50 | 14 | 122 |
| | Cat FDAO SYN | FDAO SYN | -10 | 50 | 14 | 122 |
| **Final Drive / Wheel Motors** for all models except D Series | Refer to the machine Owning and Operating Manual and/or consult your Cat dealer. | | | | | |

(continued)

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(Table 41, contd)

| Unit Rig Mining Trucks Fluid Recommendations for Ambient Temperatures[1] | | | | | | |
|---|---|---|---|---|---|---|
| Compartment or System | Oil Type and Performance Requirements | Oil Viscosities | °C | | °F | |
| | | | Min | Max | Min | Max |
| Automatic Grease System | Cat Extreme Application Grease-Arctic | NLGI 0 | -50 | 20 | -58 | 68 |
| | Cat Extreme Application Grease | NLGI 1 | -35 | 40 | -31 | 104 |
| | | NLGI 2 | -30 | 50 | -22 | 122 |
| | Cat Prime Application Grease | NLGI 2 | -20 | 40 | -4 | 104 |
| Suspension, Front, and Rear | Cat ATF-HD2 | | For all temperature ranges | | | |

(1) Refer to publication “Fluid Specifications HMS” for more details, to request contact BucyHEX-Dokumentation@cat.com.

# Underground Products

Refer to this Special Publication, “General Information for Lubricants” for important lubricant information.

## Continuous Miner

Refer to this Special Publication, “General Information for Lubricants” for important lubricant information.

This section includes, but is not limited to the following models:

- CM210 through CM445

Table 42

| Continuous Miner Fluid Recommendations for Ambient Temperatures[1] | | | | | | |
|---|---|---|---|---|---|---|
| Compartment or System | Oil Type and Performance Requirements | Oil Viscosities | °C | | °F | |
| | | | Min | Max | Min | Max |
| Hydraulics | Cat HYDO Advanced | SAE 20 | 0 | 50 | 32 | 122 |
| Gearbox | Cat Compactor Oil | ISO 220 | For all temperature ranges | | | |
| Grease points for Pins and Bushings | Cat Prime Application Grease | NLGI 2 | 0 | 50 | 32 | 122 |
| | Cat Extreme Application Grease | NLGI 1 | -35 | 40 | -31 | 104 |
| | | NLGI 2 | -30 | 50 | -22 | 122 |
| Electrical Motors | Cat Ball Bearing Grease | NLGI 2 | For all temperature ranges | | | |

(1) Refer to publication “Fluid Specifications HMS” for more details, to request contact BucyHEX-Dokumentation@cat.com.

## Roof Support Carrier, Load Haul Dump, and Face Haulage

Refer to this Special Publication, “General Information for Lubricants” for important lubricant information.

This section includes, but is not limited to the following models:

Roof Support Carrier

- SH150, SH620 through SH680

Load Haul Dump

- CL106 through CL215 and SU488 LHD

Face Haulage

- FH110, FH120, FH125D

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Table 43

| Roof Support Carrier, Load Haul Dump, and Face Haulage Fluid Recommendations for Ambient Temperatures[1] | | | | | | |
|---|---|---|---|---|---|---|
| **Compartment or System** | **Oil Type and Performance Requirements** | **Oil Viscosities** | **°C** | | **°F** | |
| | | | **Min** | **Max** | **Min** | **Max** |
| **Transmission** | Cat TDTO Cold Weather | SAE 0W-20 | -40 | 0 | -40 | 32 |
| | Cat TDTO | SAE 10W | -20 | 0 | -4 | 32 |
| | | SAE 30 | 0 | 32 | 32 | 90 |
| | | SAE 50 | 32 | No Max | 90 | No Max |
| **Hydraulic with Integral Wet Brakes** | Refer to the Operation and Maintenance Manual for Recommendations | | | | | |
| **Hydraulic System** | Cat HYDO Advanced | SAE 20 | 0 | 50 | 32 | 90 |
| **Gear Box and Wet Brake** | Refer to the Operation and Maintenance Manual for Recommendations | | | | | |
| **Axle, Wheel Ends** | Cat Gear Oil | SAE 80W-90 | For all temperature ranges | | | |
| **Winch Gearbox, Winch Planetary** [2] | Cat Compactor Oil | ISO 220 | For all temperature ranges | | | |
| **Drop Box**[3] | Cat TDTO Cold Weather | SAE 0W-20 | -40 | 0 | -40 | 32 |
| | Cat TDTO | SAE 10W | -20 | 0 | -4 | 32 |
| | | SAE 30 | 0 | 32 | 32 | 90 |
| | | SAE 50 | 32 | No Max | 90 | No Max |
| **Brake Flushing**[4] | Cat TDTO | SAE 10W | For all temperature ranges | | | |
| **Grease points for Drive Line, Pins, Bushings, Door Hinges, Tow Pin, Differential Pinion Oil Seal** | Cat Extreme Application Grease-Arctic (Semi-Synthetic) | NLGI 0 | -50 | 0 | -58 | 32 |
| | Cat Prime Application Grease | NLGI 2 | 0 | 50 | 32 | 122 |
| **Engine Drive Coupling**[5] | Cat Ball Bearing Grease | NLGI 2 | For all temperature ranges | | | |

(1) Refer to publication “Fluid Specifications HMS” for more details, to request contact BucyHEX-Dokumentation@cat.com.
(2) Applicable to Roof Support Carriers only
(3) Applicable to Machines Fitted with the C7 engine package only
(4) Applicable to Face Haulage only
(5) Applicable to Machines fitted with the 3126 engine package only

## Winterizing Lubricants

The information provided in this Special Publication, “Cold-Weather Lubricants” is applicable in general to Expanded Mining Products. In addition to this Information, Table 44 provides recommendations specific to Expanded Mining Products.

Table 44

| Winterizing Recommendations for Diesel Vehicles | | |
|---|---|---|
| **Compartment or System** | **Recommended Fluids**[1] | **Viscosity Grade** |
| **Engine Crankcase** | Cat DEO Cold Weather | SAE 0W-40 |
| **Engine Coolant** | Cat ELC<br>Cat DEAC | 50% Glycol minimum |
| **Transmission** | Cat TDTO Cold Weather | SAE 0W-20 |
| | Cat TDTO 10W | SAE 10W |

(continued)

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(Table 44, contd)

| | | |
|---|---|---|
| **Hydraulic** | Cat HYDO Advanced | SAE 10W |
| **Dropbox**(where applicable) | Cat TDTO Cold Weather<br>Cat TDTO 10W | SAE 0W-20 |
| | | SAE 10W |
| **Axle/Wheel Ends** | Cat TDTO Cold Weather | SAE 0W-20 |
| **Engine Drive Coupling** | No change from standard recommendation | |
| **Winch** | No change from standard recommendation | |

(1) For compartments with multiple fluids listed, any of the recommended fluids are acceptable.

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# Commercial Lubricants with Synthetic Base Stocks

**SMCS Code:** 1300; 1348; 7581

Some commercial lubricants are often advertised as synthetic. These commercial oils may contain a fully synthetic base stock, or a base stock that has synthetic-like properties. Some of these commercial lubricants may also contain a blend of conventional base stocks and synthetic base stocks. In any case, these synthetic oils are usually advertised as extended drain lubricants. Caterpillar does not recommend automatically extending the oil drain interval for any commercial oil, whether synthetic base stock or conventional base stock. This applies to all lubricated compartments on a Cat machine. In order to extend the oil drain interval in any lubricated compartment, refer to your machine or engine OMM, and follow the guidance given in the “Optimizing Oil Change Intervals” section of this Special Publication.

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# Re-refined Basestock Oils

**SMCS Code:** 1300; 1348; 7581

Rerefined basestock oils are acceptable for use in Cat engines **IF these oils meet the performance requirements that are specified by Cat.**

Rerefined basestock oils can be used exclusively in finished oil or in a combination with new basestock oils. The US military specifications and the specifications of other heavy equipment manufacturers also allow the use of rerefined basestock oils that meet the same criteria.

The process that is used to make rerefined basestock oil should adequately remove all wear metals and all additives that are in the used oil. Vacuum distillation and the hydrotreating of the used oil are acceptable processes that are used for producing rerefined base oil.

**Note:** Filtering is inadequate for the production of high quality rerefined basestock oils from used oil.

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# Aftermarket Oil Additives

**SMCS Code:** 1300; 1348; 7581

Caterpillar does not recommend the use of aftermarket additives in oil. It is not necessary to use aftermarket additives in order to achieve the machine's maximum service life or rated performance. Fully formulated, finished oils consist of base oils and of commercial additive packages. These additive packages are blended into the base oils at precise percentages in order to help provide finished oils with performance characteristics that meet industry standards.

There are no industry standard tests that evaluate the performance or the compatibility of aftermarket additives in finished oil. Aftermarket additives may not be compatible with the finished oil's additive package, which could lower the performance of the finished oil. The aftermarket additive could fail to mix with the finished oil. This could produce sludge. Caterpillar discourages the use of aftermarket additives in finished oils.

To achieve the best performance from Caterpillar machines, conform to the following guidelines:

- Select the proper Caterpillar oil or select commercial oil that meets the specifications designated by Caterpillar for the compartment.
- See the appropriate “Lubricant Viscosities for Ambient Temperatures” table in this publication in order to find the correct oil viscosity grade for the machine compartment.
- At the specified interval, service the engine or service the other machine compartments. Use appropriate new oil and install an appropriate new oil filter.
- Perform maintenance at the intervals that are specified in the Operation and Maintenance Manual, “Maintenance Interval Schedule”.

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# Specialty Lubricants

**SMCS Code:** 7000; 7581

Table 45

| Special-Purpose Caterpillar Lubricants | |
|---|---|
| **Item** | **Size** |
| 6V-4876 Lubricant(1) | 500 g (17.6 oz) |
| 5P-3931 Thread Compound(2) | 150 g (5.3 oz) |

(1) Recommended for use on typical components such as head bolt threads and washers.
(2) Recommended for mating connectors such as exhaust manifold studs and exhaust manifold nuts.

Table 46

| UV Visible Dyes for Leak Detection(1) | | | |
|---|---|---|---|
| **Part Number** | **Description** | **Size** | **Dosage** |
| **To detect oil leaks in engines, transmissions, hydraulic systems, etc.** | | | |
| 1U-5572 | Oil Glo 22 Additive | 28.4 gram (1 ounce) bottle | 28.4 gram (1 ounce) per 7.58 L (2 US gal) of oil |
| 1U-5573 | Oil Glo 22 Additive | 0.47 L (1 pint) bottle | 28.4 gram (1 ounce) per 7.58 L (2 US gal) of oil |
| **To detect fuel leaks or suspected fuel dilution** | | | |
| 1U-5574 | Gas Glo 32 Additive | 28.4 gram (1 ounce) bottle | 28.4 gram (1 ounce) per 37.9 L (10 US gal) of gasoline or diesel fuel |
| 1U-5575 | Gas Glo 32 Additive | 0.47 L (1 pint) bottle | 28.4 gram (1 ounce) per 37.9 L (10 US gal) of gasoline or diesel fuel |
| **To detect cooling system leaks** | | | |
| 1U-5576 | Water Glo 23 Additive | 28.4 gram (1 ounce) bottle | 28.4 gram (1 ounce) per 151.5 L (40 US gal) of water |
| 1U-5577 | Water Glo 23 Additive | 0.47 L (1 pint) bottle | 28.4 gram (1 ounce) per 151.5 L (40 US gal) of water |

(1) UV visible dyes must be used with a special glo gun in order to detect leaks. Consult your Cat dealer for availability.

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# Dry Brake Shoe Applications

**SMCS Code:** 4250-OC; 7579; 7581

This publication is a supplement to the machine Operation and Maintenance Manuals. This publication does not replace the Operation and Maintenance Manuals that are specific to the machine.

**Note:** In machine applications where U.S. Department Of Transportation (DOT) specification brake circuit actuation fluids are required, refer to the machine specific Operation and Maintenance Manual for fluid type and usage recommendations.

**Note:** DOT-3, DOT-4, and DOT-5.1 fluids are glycol based. DOT-5 fluids are silicone based.

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NOTICE
DOT-3, DOT-4, DOT-5.1, or DOT-5 fluids are not compatible with petroleum based products.

Do not mix DOT-3, DOT-4, and DOT-5.1 with DOT-5 fluids.

Do not use DOT-3, DOT-4, DOT-5.1, or DOT-5 brake circuit actuation fluids in compartments that contain friction material or wet brake systems.

Do not use DOT-3, DOT-4, DOT-5.1, or DOT-5 brake circuit actuation fluids in compartments where hydraulic oil or TO-4 fluids are recommended.

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# Dry Film Lubricant

**SMCS Code:** 7581

222-3116 Dry Film Lubricant has the following characteristics: superior lubricity, excellent adhesion to most surfaces, fast dry times and easy application.

The dry lubricant can be used for the following applications: backhoe extendable stick, blade circles for motor graders, shift mechanisms, masts for lift trucks, slides that require frequent lubrication, locks that have tumblers and applications that require a press fit. This lubricant is recommended for use on the Compact Wheel Loaders. Use the dry film lubricant for the following applications: all moving door latches, hinges, door locks, lock for the hood, hinges for the hood and throttle pedal linkage.

Use this lubricant in a well ventilated area.

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# Cold Weather Lubricants

**SMCS Code:** 1300; 1348; 7581

## Engine

NOTICE
Recommended compartment warm-up procedure must be followed. Refer to the machine Operation and Maintenance Manual. Also refer to the relevant “Lubricant Viscosities for Ambient Temperatures” tables footnotes in this Special Publication and to the “Warm-up Procedures for Machines that are used in Cold Weather - (Generic)” topic in this Special Publication.

NOTICE
Excessive engine idling time can contribute to excessive water in the crankcase oil, causing corrosion, sludge, and other problems. Excessive engine idling time can also lead to injector fouling, piston and combustion chamber deposits, corrosive damage, and increased oil consumption.

For proper selection of oil type and/or specification, refer to this Special Publication, “Engine Oil” section. Also, refer to the relevant “Lubricant Viscosities for Ambient Temperatures” tables in this Special Publication.

For the proper selection of oil viscosity grade, refer to the relevant “Lubricant Viscosities for Ambient Temperatures” table in this Special Publication. Also, refer to this Special Publication, “Lubricant Viscosities” article.

NOTICE
Not following the recommendations found in the “Lubricant Viscosities for Ambient Temperatures” tables and associated footnotes can lead to reduced performance and engine failure.

NOTICE
Do NOT use only the oil viscosities when determining the recommended oil for an engine compartment. The oil type (performance requirements) MUST also be used.

For easier cold weather starting, make sure that all the components of the engine electrical system are properly maintained. All electrical wiring and connections should be free of the following: fraying, damaged insulation, and corrosion. Batteries should be kept fully charged and warm. The batteries and the battery cables need to be the proper size for the application.

Various starting aids are available to assist with cold engine starts in low temperature conditions. Follow the recommendations that are provided by the manufacturer of the starting aid. Refer to the “Aftermarket Products and Warranty” article in the “ Warranty Information” section of this special publication.

Additional information on cold-weather operation is available in Special Publication, SEBU5898, “Cold Weather Recommendations For All Caterpillar Equipment”. This publication is available from your Cat dealer.

Additionally, for more information on cold-weather operation, refer to this Special Publication, “Fuel Specifications” section. Also refer to this Special Publication, “Cooling System Specifications” (Maintenance Section).

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Before attempting to start the engine, make sure that the oil in the engine is fluid enough to flow. Check the oil by removing the dipstick. If the oil will drip from the dipstick, then the oil should be fluid enough to allow the engine to start. Do not use oil that has been diluted with kerosene. Kerosene will evaporate in the engine. Evaporation will cause the oil to thicken. Kerosene will cause swelling and softening of the silicone seals. Kerosene will dilute the oil additives. Dilution of the oil additives will reduce the oil performance, and reduce the engine protection that the additives provide. If your machine is equipped with a gasoline starting engine (earlier machine), make sure that the oil is fluid enough to flow.

If the viscosity of the oil is changed for colder weather, also change the filter element. If the filter is not changed, the filter element and the filter housing can become a solid mass. After you change the oil, operate the engine to circulate the thinner oil.

When you start a cold-soaked engine or when you operate an engine in ambient temperatures that are below −18°C (0°F), use base oils that can flow in low temperatures. These multigrade oils have lubricant viscosity grade of SAE 0W or of SAE 5W. An example of viscosity grade is SAE 5W-40.

When you start a cold-soaked engine or when you operate an engine in ambient temperatures that are below −30°C (−22°F), use a synthetic basestock multigrade oil. The oil should have a lubricant viscosity grade of SAE 0W or SAE 5W. Use an oil with a pour point that is lower than −40°C (−40°F).

**Note:** Use the highest oil viscosity grade that is allowed for the ambient temperature when you start the engine. If a different oil viscosity grade is specified in “Lubricant Viscosities for Ambient Temperatures”, use the viscosity grade that is specified in the table. **In arctic applications, a properly sized engine compartment heater is recommended, and use a higher viscosity grade oil.** Refer to the “Lubricant Viscosities” article in this Special Publication for further details.

**Note:** Cold-soaked starts occur when the engine has not been operated for a time. The oil becomes more viscous due to cooler ambient temperatures. Supplemental heat is recommended for cold-soaked starts that are below the minimum ambient temperatures listed in the “Lubricant Viscosities for Ambient Temperatures” tables. Supplemental heat may be required for cold-soaked starts that are above the minimum temperature that is stated, depending on the parasitic load and other factors.

NOTICE
**Engines that use fluid or pan heaters, or heated enclosures, or are kept running under load, etc. can, and generally should use higher viscosity oil.** The “Lubricant Viscosities for Ambient Temperatures” tables (Maintenance Section) **Minimum** viscosity for ambient temperature recommendations are for cold-soaked conditions. Use the highest viscosity oil that is allowed for the ambient temperature at startup. **BUT, under continuous usage (multiple shifts per day),** and/or when using **fluid or pan heaters**, use a higher viscosity oil than the minumum recommended viscosity for cold-soaked starting conditions. The higher viscosity oil will maintain the highest possible oil film thickness. Refer to the “Lubricant Viscosities for Ambient Temperatures” tables and the table footnotes for exceptions.

**Example:** The oil viscosity recommended for use in Cat diesel engines for cold-soaked starts at −40 °C (−40 °F) is multigrade oil of the SAE 0W viscosity grade (SAE 0W-30). If the diesel engine is run continuously, SAE 15W-40 viscosity grade diesel engine oil can be used and is generally the preferred oil viscosity in this situation.

NOTICE
If ambient conditions warrant, a higher viscosity oil of the recommended specification for a given compartment may need to be installed in order to provide adequate film thickness.

## Non-Engine Machine Compartments

NOTICE
Recommended compartment warm-up procedure must be followed. Refer to the machine Operation and Maintenance Manual. Also refer to the relevant “Lubricant Viscosities for Ambient Temperatures” tables footnotes in this Special Publication and to the “Warm-up Procedures for Machines that are used in Cold Weather - (Generic)” topic in this Special Publication.

For the proper selection of oil type and/or specification, refer to this Special Publication, “Lubricant Specifications” section. Also, refer to the relevant “Lubricant Viscosities for Ambient Temperatures” tables (Oil Type and Specification column) and table footnotes in this Special Publication.

For the proper selection of oil viscosity grade, refer to this Special Publication, “Lubricant Viscosities for Ambient Temperatures” tables. Also, refer to this Special Publication, “Lubricant Viscosities” article.

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NOTICE
Not following the recommendations found in the “Lubricant Viscosities for Ambient Temperatures” tables and associated footnotes can lead to reduced performance and compartment failure.

If the viscosity of the oil is changed for colder weather, also change the filter element. If the filter is not changed, the filter element and the filter housing can become a solid mass. After you change the oil, operate the engine to circulate the thinner oil.

**Note:** Use the highest oil viscosity grade that is allowed for the ambient temperature when you start the machine. If a different oil viscosity grade is specified in the “Lubricant Viscosities for Ambient Temperatures” table, use the viscosity grade that is specified in the table. **In arctic applications, a properly sized engine compartment heater is recommended, and use a higher viscosity grade oil.** Refer to the “Lubricant Viscosities” article in this Special Publication for further details.

**Note:** Cold-soaked starts occur when the machine has not been operated for a time. The oil becomes more viscous due to cooler ambient temperatures.

NOTICE
**Machines that use fluid or pan heaters, or heated enclosures, or are kept running under load, etc. can, and generally should use higher viscosity oil.** The “Lubricant Viscosities for Ambient Temperatures” tables (Maintenance Section) **“Minimum”** viscosity for ambient temperature recommendations are for cold-soaked conditions. Use the highest viscosity oil that is allowed for the ambient temperature when you start the machine **BUT,** under **Continuous Usage (Multiple Shifts/Day),** and/or when using **fluid or pan heaters**, etc., use a higher viscosity oil, **NOT** the oil with the minumum recommended viscosity for cold-soaked starting conditions. The higher viscosity oil will maintain the highest possible oil film thickness. Refer to the “Lubricant Viscosities for Ambient Temperatures” tables and the table footnotes for exceptions.

NOTICE
Some machine compartments do not allow the use of SAE 0W, SAE 5W or certain other viscosity grade oils. Refer to the tables for “Lubricant Viscosities for Ambient Temperatures” that are in this Special Publication.

NOTICE
If ambient conditions warrant, a higher viscosity oil of the recommended specification/category for a given compartment may need to be installed in order to provide adequate film thickness.

NOTICE
Recommended compartment warm-up procedure must be followed. Refer to the machine Operation and Maintenance Manual. Also refer to the relevant “Lubricant Viscosities for Ambient Temperatures” tables footnotes in this Special Publication and to the “Warm-up Procedures for Machines that are used in Cold Weather - (Generic)” topic in this Special Publication.

## Warm-Up Procedures for Machines that are used in Cold Weather (Generic)

**Note:** For recommendations that are specific to your machine, refer to the Operation and Maintenance Manual for your machine.

After the engine is warm, warm up the other systems. Start with the hydraulic system. Run the engine at less than one-third throttle and slowly move the control lever to lift the attachment. Initially, lift the control lever for a few centimeters (inches). Lower the attachment slowly. Continue the following sequence: raising, lowering, extending, and retracting. Extend the travel during each cycle. This operation must be performed for all hydraulic circuits. Alternate between all the attachments.

Exercise the transmission and the power train. If you cannot move the control for the transmission, perform the following steps:

- Engage the parking brake or apply the parking brake.
- Run the engine slightly above LOW IDLE.
- Shift the transmission several times from FIRST GEAR FORWARD to FIRST REVERSE.

Release the brake. Move the equipment forward and backward for several meters (yards). Exercise the machine for several minutes.

To reduce the total warm-up time, start exercising the entire machine before you complete the hydraulic warm-up time.

Operate under a light load until the systems reach normal operating temperatures.

If the engine temperature is not high enough, enclose the engine and block the radiator. A thermostat that opens at a higher temperature will not increase the engine temperature if the engine is not under load.

To prevent seal damage and gasket damage, keep the pipe for the engine crankcase breather clear of blockage.

In extreme conditions, use a canvas over the engine compartment. Heat the engine area with a space heater. Heating will aid in starting the engine. Extending the canvas over the hydraulic components will provide initial warming of the components. **Follow all applicable safety guidelines.**

Running the engine at low idle will not keep the hydraulic systems warm.

Cold-weather operations require more time for completion than other operations. The extra time that is spent in properly caring for the equipment can prolong the life of the equipment. Extra care is especially helpful in extreme conditions. Longer equipment life will decrease overall cost.

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# S·O·S Services Oil Analysis

**SMCS Code:** 1000; 1348; 3080; 4070; 4250; 4300; 5095; 7000; 7542; 7581

NOTICE
These recommendations are subject to change without notice. Consult your local Cat dealer for the most up to date recommendations.

**Note:** DO NOT USE ONLY THIS SPECIAL PUBLICATION AS A BASIS FOR DETERMINING OIL DRAIN INTERVALS.

Evaluation of oils to determine their condition as they are used in the engines or machine compartments is a powerful tool to indicate the engine or component performance and to support durability and reliability that has been designed into these engines and machine components.

Caterpillar has developed a maintenance management tool that evaluates oil degradation and detects the early signs of wear on internal components. The Cat tool for oil analysis is called S·O·S Oil Analysis and the tool is part of the S·O·S Services program. S·O·S Oil Analysis divides oil analysis into four categories:

- Component wear: elemental analysis of the oil evaluates wear metals and contamination in the oil. Wear metals analysis allows the evaluation of wear of lubricated components or engines.
- Oil condition: determines if the oil has degraded due to chemical or thermal stresses. This analysis includes oxidation, nitration, sulfation, viscosity, soot and if needed total acid number and/or total base number.
- Oil contamination: testing of harmful contaminants that may have entered the oil including dirt, water, fuel and coolants. Oil contamination can cause wear and damage of engine and lubricated components.
- Oil identification: this analysis ensures that the appropriate quality of oil is used and that the right oil is used in the right compartment. The wrong oil in a compartment can severely damage major components.

These four types of analysis are used to monitor the condition of your equipment, and to help you identify potential problems. A properly administered S·O·S Services Oil Analysis program will reduce repair costs and the program will lessen the impact of downtime.

The S·O·S Oil Analysis program uses a wide range of tests to determine the condition of the oil and the condition of the lubricated compartment. Guidelines that are based on experience and a correlation to failures have been established for these tests. A trained person at your Cat dealership should make the final analysis.

Oil analysis is one of the diagnostic tools to determine engine health. Oils that are within the limits given by the guidelines may not indicate all engine health issues. Under certain conditions, including, but not limited to severe operating conditions, oils that are within the limits given by the guidelines may require changing early.

**Note:** Cooling system problems will also reduce the life of engines, transmissions, and hydraulic systems. S·O·S Coolant Analysis together with S·O·S Oil Analysis provide a complete and accurate method for monitoring the health of all machine systems. Refer to the S·O·S Coolant Analysis information in this publication. A properly administered S·O·S Services program will reduce repair costs and lessen the impact of downtime.

Refer to the “Contamination Control” article in this Special Publication for recommended fluid cleanliness targets.

**Note:** Most oil analysis programs do not detect larger particles in the oil sample. Some failure modes only produce larger particles. Oil analysis alone will not always detect an impending failure. Oil filters should be sectioned and inspected for the presence of visible particles.

The engine oil consumption must be measured and recorded. A significant increase in oil consumption can indicate a problem with cylinder pack deposits or components. Additionally, oil additions dilute wear metals and other contaminants. Oil analysis results may become inaccurate.

Consult your Cat dealer for complete information and assistance about the S·O·S Oil Analysis program.

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## Obtaining S·O·S Oil Samples

Before you obtain an S·O·S oil sample, operate the machine until the oil is warm and the oil is well circulated. Then obtain the S·O·S oil sample.

In order to obtain a good oil sample, do not take the oil sample from the drain stream. The drain stream method can allow a stream of dirty oil from the bottom of the compartment to contaminate the sample. Likewise, never dip an oil sample from an oil container or pour a sample from a used filter.

NOTICE
Always use a designated pump for oil sampling, and use a separate designated pump for coolant sampling. Using the same pump for both types of samples may contaminate the samples that are being drawn. This contaminate may cause a false analysis and an incorrect interpretation that could lead to concerns by both dealers and customers.

There are two ways to obtain S·O·S oil samples. The following methods are listed in the order that is preferred:

- Use an in-line sampling valve for pressurized oil systems.
- Use a sampling gun that is inserted into the sump.

Use of the in-line sampling valve is the preferred method. This method provides samples that are less likely to be contaminated. Whenever you obtain the samples, obtain the samples from the same point. The samples will be more representative of the oil that is in the system.

Normally, the oil sample is taken at low idle. If the flow rate is too low, increase engine speed to obtain the oil sample.

In-line sampling valves cannot be used on nonpressurized oil systems such as differentials and final drives. Use of the sampling gun is the preferred method for nonpressurized oil systems.

Refer to the Operation and Maintenance Manual, “Maintenance Interval Schedule” for the proper interval.

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# Oil Sampling Interval

**SMCS Code:** 1000; 3000; 4000; 4050; 4250; 4300; 5050; 7000; 7542

Take the oil samples as close as possible to the standard intervals. To receive the full value from S·O·S Oil Analysis, establish a consistent trend of data. To establish a pertinent history of data, perform consistent oil samplings that are evenly spaced.

General recommendations for oil sampling intervals for engines and machine compartments are given in Table 47 in this Section.

**Note:** Refer to the Operation and Maintenance Manual that came with your machine for the recommended oil sampling intervals for each compartment. Severe applications may require a more frequent oil sampling interval.

For other acceptable oil types and specifications, refer to the “Lubricant Viscosities for Ambient Temperatures” Tables in this Special Publication.

Refer to the Operation and Maintenance Manual that came with your machine for the recommended oil change intervals for each compartment.

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Table 47

| Compartment | Recommended Sampling Interval (1) | Sampling Valve | Oil Type |
|---|---|---|---|
| Engine | 250 Hours | Yes | Cat DEO<br>Cat DEO-ULS |
| Transmission | 500 Hours | Yes | Cat TDTO<br>Cat TDTO-TMS |
| Hydraulics | 500 Hours | Yes | Cat HYDO Advanced |
| Differential and Final Drive | 500 Hours | No | Cat TDTO<br>Cat FDAO |

(1) These sampling intervals are general. Refer to your engine or machine OMM for sampling intervals specific to your engine or machine.

Consult your Cat dealer for complete information and assistance in establishing an S·O·S Services program for your equipment.

## More Frequent S·O·S Sampling Improves Life Cycle Management

Recommended S·O·S sampling intervals have been determined by Caterpillar to maximize engine and component life. Table 47 in this Section provides the generally recommended sampling intervals. However, in severe service applications, more frequent oil sampling is recommended. Severe service for lubricated compartments occurs at high loads, in high temperatures, and in dusty conditions. If any of these conditions exist, work with your Cat Dealer to determine an appropriate sampling interval. In some cases, this may be half the normal recommended interval or even more frequently. These additional samples will increase the chance of detecting a potential failure and reducing unexpected down time.

### Determining Optimum Oil Change Intervals

The recommended oil drain intervals for your machine are published in the Cat Operation and Maintenance Manual (OMM). These oil drains are for typical applications where the listed practices are followed:

- Use recommended oils, as described in this document or the OMM
- Use of recommended fuels, as described in this document or the OMM
- Use of Cat oil filters
- Use of Caterpillar recommended maintenance

In the vast majority of applications, the recommended oil drain intervals are designed to provide excellent protection for your machine. In a few specialized applications, there may be a need for shorter oil drains. The need for a shorter oil drain will be identified through oil sampling and oil analysis.

Improvements in product design and improvements in lubricant performance have created the opportunity for extended oil drains in all lubricated compartments. Some of the extended oil drains are reflected in current OMM recommendations. The common requirements for these extended oil drains are:

- Use of Cat branded lubricants
- Use of Cat filters
- Use of CAt S·O·S Services oil analysis

These three requirements lower the risk of an extended oil drain and enable the longest oil drains. Your Cat dealer can provide you with the best lubricants for extended oil drains.

With these improvements to product design and lubricant performance, extended oil drains are a proven way to lower maintenance costs and improve profitability. A properly managed extended oil drain program allows the full life designed into the machine and supports recommended component overhauls. Extended oil drains do allow you to obtain all the life designed into Cat lubricants and reduce waste.

The process for extending engine oil drains is available in the brochure “Optimizing Oil Change Intervals”PEHJ0192. This process is basically the same for all lubricated compartments on a Cat machine. Work with your Cat dealer to select the proper high-performance Cat lubricant for your extended oil drain program. Your Cat dealer has additional resources to guide you in the extended oil drain process.

### Cat Lubricants for Extended Oil Drains

All Cat lubricants are high performance products, and are recommended for extended oil drains. Within each group of Cat lubricants, there are products available at different performance levels. Your Cat dealer has the list of Cat lubricants that are available in your region of the world. Work with your Cat dealer to select the best Cat lubricant for your extended oil drain program.

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## Fuel Sulfur Impact on Engine Oil

The use of Cat S·O·S Services oil analysis is recommended for determining oil life.

For Cat machines operating under emissions regulations such as US EPA Tier4, EU Stage V or other emission regulations controls, the maximum fuel sulfur level permitted by regulations in the US is 0.0015% or 15 ppm. Maximum permitted fuel sulfur levels in other emissions regulated countries may vary from .0015% or 15 ppm slightly and must be followed.

Cat diesel engines that do not use emissions reduction devices (do not operate under emissions regulations), or meet Tier1, Tier2, or Tier3 emissions levels may run on diesel fuels that exceed 0.0015% sulfur. However, using higher sulfur fuels may shorten the oil change interval.

**In order to help protect your engine and optimize oil drain intervals for engine applications and duty cycles**, it is important to use SOS Services oil analysis to determine if the sulfur has degraded the oil. **Use Cat S·O·S Services oil analysis per the following general guidelines:**

- Recommended normally
- For fuel sulfur level up to 0.05 percent (500 ppm), no additional sampling is required. Follow the recommended oil sampling in the machine OMM
- For fuel sulfur level of > 0.05 percent to 0.5 percent (500 ppm to 5000 ppm), SOS services oil analysis is strongly recommended to determine oil drain intervals. Sample the oil every 250 hours until a trend is established, then sample as needed.
- For fuel sulfur level of > 0.50 percent (>5000 ppm), SOS services oil analysis is required to determine oil drain intervals. Sample oil every 125 hours until a trend is established, then sample as needed.

These recommendations apply for Cat DEO-ULS and DEO oils.

**Note:** Engine operating conditions play a key role in determining the effect that fuel sulfur will have on engine deposits and on engine wear. Refer to the Diesel Fuel Chapter in this Special Publication and consult your Cat dealer for guidance when fuel sulfur levels are above 0.2% (2000 ppm) Base the oil change interval on the oil analysis. Ensure that the oil analysis includes the condition of the oil and a wear metal analysis.

Consult a trained S·O·S Services analyst when making oil drain decisions based on oil sample results.

NOTICE
Depending on application severity and localized environmental conditions, and also depending on maintenance practices, operating Direct Injection (DI) diesel engines and operating PC (Precombustion Chamber) diesel engines on fuel with sulfur levels over 0.1 percent (1000 ppm) may require significantly shortened oil change intervals in order to help maintain adequate wear protection. Refer to this Special Publication, “Fuel Specifications” section, “Diesel Fuel Sulfur” topic for additional information.

**Note:** For PC (Precombustion Chamber) diesel engines, which are mainly 1990 and older engines, the minimum new oil TBN must be 20 times the fuel sulfur level.

For fuel and coolant analysis and use of SOS services, refer to the Diesel Fuel and Coolants chapters in this Special Publication.

**Note:** Engine operating conditions play a key role in determining the effect that fuel sulfur will have on engine deposits and on engine wear. Consult your Cat dealer for guidance when fuel sulfur levels are above 0.2% (2000 ppm).

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# Lubricating Grease

**SMCS Code:** 0645; 1000; 7000; 7581

NOTICE
These recommendations are subject to change without notice. Consult your local Cat dealer for the most up to date recommendations.

**Note:** Non-Cat commercial greases are as a group second choice greases. Within this grouping of second choice greases there are tiered levels of performance.

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids and greases.

Caterpillar provides a family of grease products that vary in performance from moderate to high performance. These greases service the entire line of Cat products in the wide variety of climates throughout the world. From this variety of Cat grease products, you can find a Cat grease that will meet or exceed the performance requirements for almost every machine that is produced by any Original Equipment Manufacturer (OEM), and for almost every machine application or equipment application.

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Before selecting a grease product, the performance requirements must be determined. Consult the grease recommendations in your Operation and Maintenance Manual. Also, consult your Cat dealer for a list of greases that have the performance specifications and the available container sizes.

This grease chapter refers to the NLGI grade of the grease products offered by Caterpillar. The National Lubricating Grease Institute (NLGI) classifies the greases per their consistency or hardness according to “ASTM D4950” and “SAE J310”. The higher the NLGI grade the more firm the grease. The NLGI grade is an important classification for greases and it should be used along with other properties such as stability, corrosion resistance, viscosity, etc.

**Note:** Always choose grease that meets or exceeds the recommendations that are specified by the equipment manufacturer for the application.

Some work sites require the use of a single grease to fulfill the needs of all the equipment. Always choose a grease that meets or exceeds the requirements of the most demanding application. Remember that commercial products which barely meet the minimum performance requirements can be expected to produce the minimum life for the parts. It is not recommended to purchase grease products based on the lowest cost as the only consideration. Instead, use the grease that yields the lowest **total** operating cost. This cost should be based on an analysis that includes the improved product life, costs of parts, labor, downtime, and the cost of the amount of grease that is required.

**Note:** Certain Cat grease products are formulated with Molybdenum disulfide ($MoS_2$ or “Moly”). The Moly used in Cat greases is of a technical fine grade. This size meets the special requirements of some rolling element bearings, joints, and other moving components of tight clearances.

**Note:** When the grease in a joint is changed from one type of grease to another or a grease from a different supplier is to be used, the general recommendation is to purge all of the old grease from the joint. Some greases are not chemically compatible. Consult your supplier in order to determine if the greases are compatible.

**If in doubt, Purge!**

**Note: All Cat greases are “chemically” compatible with each other. Mixing of Cat greases will not result in an adverse chemical reaction. However, mixing of Cat greases might result in reduced performance.**

## Moderate Applications

The Caterpillar family of greases includes several which are formulated for moderate applications of medium loads and speeds. These greases are formulated to provide optimal performance for these applications.

## Cat Utility Grease

Cat Utility Grease is formulated for use in applications that have a low severity to a medium severity and moderate temperatures. Cat Utility Grease is a National Lubricating Grease Institute (NLGI) grade 2 grease. This grease is made with petroleum base oil and a lithium complex thickener.

**Note:** If the application calls for a utility grease and Cat Utility Grease is not available, consult the grease data sheets. Use a substitute that meets or exceeds the performance characteristics of Cat Utility Grease.

## Cat White Assembly Grease

Cat White Assembly Grease is formulated for use in applications that have low to medium severity and moderate temperatures. This grease has been made extra tacky in order to hold gaskets, O-rings, and needle bearings to aid in the assembly of engines, transmissions, and other components. Cat White Assembly Grease is an NLGI grade 2 grease. This grease is made with a petroleum base oil and a lithium complex thickener.

## Cat Prime Application Grease

Cat Prime Application Grease is formulated for use in applications with low severity to high severity at moderate temperatures. Cat Prime Application Grease is recommended for heavily loaded pin joints and high impact applications in machines such as track-type tractors, backhoe loaders and skid steer loaders.

Cat Prime Application Grease is an NLGI grade 2 grease. This grease is made with petroleum base oil and a lithium complex thickener. This grease has 3% Molybdenum diSulfide ($MoS_2$ or “Moly”).

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